

Tan Yong Jie

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Robotics Systems Engineering student at SIT, year 2. Shipped PoC systems to industry stakeholders in computer vision and embedded systems. Built and deployed autonomous navigation systems in robotics coursework. Two-time WorldSkills Singapore Medallion of Excellence winner across three national competitions.

COMPETITION

Worldskills Singapore - Mechatronics

Medallion of Excellence (2020 & 2021)

Three-time competitor (2018, 2020, 2021), winning the Medallion of Excellence twice. Worked in a 2-person team to design, assemble, and program modular industrial automation systems under competition conditions, covering PLC, HMI, pneumatics, and robotics.

PROJECTS

Omnidirectional Perception System (OPS) - Technical Lead (Software)

Jan - Mar 2026

ROS2 Humble, Python, OpenCV, Tesseract OCR, pyzbar, SQLite, Raspberry Pi | [GitHub](#)

- Built and delivered an omnidirectional barcode scanning system for an industry client. It replaced manual scanning across inbound and sorting operations, cutting human error at the source.
- Designed SmartMatcher engine fusing OCR and barcode data per item, handling degraded and misoriented barcodes without failure.
- Tuned the Pi camera pipeline until all 3 cameras scanned within 2.5s. Down from roughly 6s in the first build.

Autonomous Maze Escape Robot

Sep - Dec 2025

ROS2 Humble, Python, SLAM Toolbox, Nav2, Gazebo, Turtlebot3 | [GitHub](#)

- Built an autonomous maze escape system, navigating TurtleBot3 from random spawn to exit. Initially targets a fixed far goal, then switches to max LiDAR range as a goal when the forward path is clear.
- Built LiDAR exit detection requiring 80%+ forward readings over 5s. Zero false positives.
- Wrote a 4-direction recovery system for stuck scenarios. Achieved 95%+ success rate.

Autonomous Indoor Navigation - Team Leader (Software)

May - Aug 2025

ROS1 Noetic, Python, RTAB-Map, Navigation Stack, RViz, Jetson Nano, AgileX LIMO | [GitHub](#)

- Led a 6-person software team to navigate AgileX LIMO across all 8 randomized plots in a 4,500x3,990 mm shared arena.
- Implemented custom recovery behavior (costmap clearing, in-place rotation, and reverse escape) using MoveBaseAction and /move_base/make_plan.
- Fused depth camera, 2D LiDAR, IMU, and wheel encoders for AMCL localization and real-time SLAM. No drift-induced failures across the full arena.

EXPERIENCE

Singapore Institute of Technology - Student Technologist

Aug 2025 to Apr 2026

- Migrated 16 AgileX LIMO robots from Jetson Nano to Orin Nano to support ROS2 Humble deployment across university teaching labs.
- Validated full stack compatibility across chassis drivers, LiDAR, and depth cameras on each unit. No regressions post-migration.
- Supported 60 students across teaching labs, with zero downtime on all 16 robots.

CERTIFICATIONS

Certified SOLIDWORKS Associate (CSWA) — Dassault Systèmes

EDUCATION

B.Eng (Hons) Robotics Systems Engineering (Singapore Institute of Technology)

Sep 2024 - Present (Expected 2028)

Diploma in Robotics & Mechatronics Engineering (Nanyang Polytechnic)

Apr 2019 - Mar 2022

Director's List (3 semesters) · WorldSkills Singapore Mechatronics 2020 & 2021

LANGUAGES & TECHNOLOGIES

Robotics & Navigation: ROS1, ROS2, Nav2, SLAM Toolbox, RTAB-Map, TurtleBot3, AgileX LIMO

Computer Vision: OpenCV, Tesseract OCR, pyzbar, LiDAR, Depth Camera

Programming: Python, C, C++ (STM32, Parallax embedded work), Linux, Git

Embedded & Industrial: STM32, Raspberry Pi, Jetson Nano, TwinCAT PLC, Panasonic PLC/HMI (LD, SFC)

Tools: RViz, Gazebo, SQLite

Languages: English (fluent), Mandarin (spoken)